Symbolic Projection Notes for $B_{c1} \to B_c^{(*)}\gamma$

Working notes

July 11, 2026

1 Scope

These notes collect the compact FeynCalc projection results for the heavy-heavy channels

$$B_{c1}(6743) \to B_c\gamma, \qquad B_{c1}(6743) \to B_c^*\gamma, \qquad B_{c1}(6750) \to B_c\gamma, \qquad B_{c1}(6750) \to B_c^*\gamma.$$

The calculation is structurally different from the D_s and B_s light-cone sum rules. In the B_c system both valence constituents are heavy, so the leading radiative OPE is a hard-photon heavy-heavy calculation. There is no light spectator line to be replaced by photon distribution amplitudes, and no $\chi_s\langle\bar{s}s\rangle$ type term.

2 Currents and Propagators

For $B_c^+ \sim c\bar{b}$, we use

$$j_5 = \bar{b}\, i\gamma_5 c, \qquad j_\rho^V = \bar{b}\,\gamma_\rho c,$$

and the two axial basis currents

$$J_\mu^A = \bar{b}\,\gamma_\mu\gamma_5 c, \qquad J_\mu^B = \frac{i}{m_b + m_c}\,\bar{b}\,\sigma_{\mu\alpha}P^\alpha\gamma_5 c, \qquad P = p + q.$$

The physical currents are

$$J_{1\mu} = \sin\theta\, J_\mu^A + \cos\theta\, J_\mu^B, \qquad J_{2\mu} = \cos\theta\, J_\mu^A - \sin\theta\, J_\mu^B.$$

This mixing is used at the invariant-amplitude level. The OPE is first derived for the basis currents because the two Dirac traces are independent; the physical amplitudes are then

$$\widehat{\Pi}_1 = \sin\theta\,\widehat{\Pi}_A + \cos\theta\,\widehat{\Pi}_B, \qquad \widehat{\Pi}_2 = \cos\theta\,\widehat{\Pi}_A - \sin\theta\,\widehat{\Pi}_B.$$

For the pseudoscalar final state,

$$g_i = \frac{m_b + m_c}{m_{B_c}^2 f_{B_c}\, m_i f_i} \exp\left(\frac{m_i^2}{M_1^2} + \frac{m_{B_c}^2}{M_2^2}\right)\widehat{\Pi}_i^{(P)},$$

while for the vector final state,

$$g_i^* = \frac{1}{m_{B_c^*} f_{B_c^*}\, m_i f_i} \exp\left(\frac{m_i^2}{M_1^2} + \frac{m_{B_c^*}^2}{M_2^2}\right)\widehat{\Pi}_i^{(V)}.$$

The updated $B_c\gamma$ script applies this rotation with physical two-point residues f_1 and f_2. The vector $B_c^*\gamma$ entries remain diagnostic until their tensor-basis normalization is audited.

The free heavy-quark propagator in momentum space is

$$S_Q^{(0)}(k) = \frac{i(\not{k} + m_Q)}{k^2 - m_Q^2 + i0}, \qquad Q = c, b.$$

The hard photon is included by inserting the electromagnetic vertex on either heavy line:

$$S_Q^{(\gamma)}(k,k+q) = S_Q^{(0)}(k+q)\, ie_Q\gamma_\nu\epsilon^\nu\, S_Q^{(0)}(k).$$

For the $\bar{b}$ line it is convenient to use the effective antiquark charge $e_{\bar{b}} = +e/3$, while $e_c = +2e/3$. Equivalently, one may keep $e_b = -e/3$ and track the orientation of the antiquark line. The scripts use $e_{\bar{b}}$ explicitly.

Thus, yes: this is the same heavy-heavy propagator logic as in a B_c-mixing sum-rule calculation. The important difference from the strange-sector radiative calculation is that we do not insert a light-quark photon DA. If we later include power corrections, the natural heavy-heavy correction is the background-gluon expansion of the heavy propagators, e.g. gluon-condensate terms, not strange-quark condensate photon DAs.

The submitted B_c-mixing work uses the same local basis,

$$J_\mu^A = \bar{b}\gamma_\mu\gamma_5 c, \qquad J_\mu^B = \frac{i}{m_b+m_c}\bar{b}\sigma_{\mu\alpha}p^\alpha\gamma_5 c,$$

with $m_b = 4.18$ GeV, $m_c = 1.27$ GeV, and the final working window

$$7\ \text{GeV}^2 \le M^2 \le 9\ \text{GeV}^2, \qquad 53\ \text{GeV}^2 \le s_0 \le 55\ \text{GeV}^2.$$

Those are useful starting values for the heavy-heavy OPE in the radiative problem. They are not yet automatically the final radiative-transition windows; the final windows must be fixed by the three-point sum-rule stability, pole dominance, and convergence tests.

3 Step-by-Step Sum-Rule Construction

This section is written as a checklist for reproducing the calculation.

1. Start from basis currents. The physical currents J_1, J_2 are mixed combinations of J_A, J_B. We derive the OPE for J_A and J_B separately only because the Dirac traces are simpler. The physical amplitudes are obtained afterward through the $\sin\theta, \cos\theta$ rotation.

2. Write the correlators. For $B_c\gamma$,

$$\Pi_\mu^{(5,K)}(p,q) = i\int d^4x\, e^{ipx}\langle\gamma(q,\epsilon)|T\{\bar{b}(x)i\gamma_5 c(x)\, J_{K\mu}^\dagger(0)\}|0\rangle, \qquad K = A, B.$$

For $B_c^*\gamma$,

$$\Pi_{\mu\nu}^{(V,K)}(p,q) = i\int d^4x\, e^{ipx}\langle\gamma(q,\epsilon)|T\{\bar{b}(x)\gamma_\nu c(x)\, J_{K\mu}^\dagger(0)\}|0\rangle.$$

3. Contract the heavy fields. There are two hard-photon diagrams: photon emission from the c line and photon emission from the $\bar{b}$ line. The free heavy propagator is

$$S_Q^{(0)}(k) = \frac{i(\not{k}+m_Q)}{k^2-m_Q^2+i0},$$

and the photon insertion is

$$S_Q^{(\gamma)}(k,k+q) = S_Q^{(0)}(k+q)\, ie_Q\not{\epsilon}\, S_Q^{(0)}(k).$$

The charge convention in the scripts is $e_c = +2/3$ and $e_{\bar{b}} = +1/3$. No photon DA or light-quark condensate term appears in this heavy-heavy leading baseline.

4. Evaluate and project the traces. For $B_c\gamma$ project onto

$$S_{\mu\nu} = p_\nu q_\mu - (p\cdot q)g_{\mu\nu}, \qquad \mathcal{P}^{\mu\nu} = \frac{S^{\mu\nu}}{2(p\cdot q)^2}, \qquad S_{\mu\nu}\mathcal{P}^{\mu\nu} = 1.$$

For $B_c^*\gamma$ the current diagnostic tensor is

$$V_{\mu\nu\rho} = p_\nu \epsilon_{\mu\rho p q} - (p\cdot q)\epsilon_{\mu\nu\rho p}, \qquad V_{\mu\nu\rho}V^{\mu\nu\rho} = -4p^2(p\cdot q)^2.$$

The vector-channel result still needs matching to the physical $1^+ \to 1^-\gamma$ amplitude basis.

5. Reduce to denominator structures. For photon emission from the charm line use

$$D_1 = (k+q)^2 - m_c^2, \quad D_2 = k^2 - m_c^2, \quad D_3 = (k-p)^2 - m_b^2.$$

For photon emission from the anti-bottom line use

$$D_1 = k^2 - m_c^2, \quad D_2 = (k-p)^2 - m_b^2, \quad D_3 = (k-p+q)^2 - m_b^2.$$

The triangle pieces contain the full three-denominator product. Terms where a numerator cancels one denominator become two-denominator/contact terms and must be included or shown to vanish.

6. Borel match and rotate. The true variables are p^2 and $P^2 = (p+q)^2$, with

$$p\cdot q = \frac{P^2 - p^2}{2}.$$

The pilot numerical calculation uses the diagonal choice $M_1^2 = M_2^2 = 2M^2$, leading to the practical exponential

$$\exp\left[\frac{m_i^2 + m_f^2}{2M^2}\right].$$

After obtaining $\widehat{\Pi}_A, \widehat{\Pi}_B$, the physical combinations are

$$\widehat{\Pi}_{6743} = \sin\theta\,\widehat{\Pi}_A + \cos\theta\,\widehat{\Pi}_B, \qquad \widehat{\Pi}_{6750} = \cos\theta\,\widehat{\Pi}_A - \sin\theta\,\widehat{\Pi}_B.$$

For the $B_c\gamma$ widths we use

$$\Gamma(1^+ \to 0^-\gamma) = \frac{\alpha_{\rm em}}{3} g^2 E_\gamma^3, \qquad E_\gamma = \frac{m_i^2 - m_f^2}{2m_i}.$$

4 $B_{c1} \to B_c\gamma$: E1 Projection

For the pseudoscalar final state we project onto

$$S_{\mu\nu} = p_\nu q_\mu - (p\cdot q)g_{\mu\nu}, \qquad \mathcal{P}_{\mu\nu} = \frac{S_{\mu\nu}}{2(p\cdot q)^2}.$$

FeynCalc gives the normalization check

$$S_{\mu\nu}\mathcal{P}^{\mu\nu} = 1.$$

We denote

$$p^2 = p_2, \qquad p\cdot q = pq.$$

After rewriting loop scalar products in terms of inverse denominators and setting those denominators to zero, the projected triangle-core numerators are

$$
\begin{aligned}
C_A^{(c)} &= -4i\,m_c + \frac{4i\,m_b m_c^2}{pq} - \frac{4i\,m_c^3}{pq}, \\
C_A^{(\bar b)} &= -4i\,m_b + \frac{4i\,m_b^3}{pq} - \frac{4i\,m_b^2 m_c}{pq},
\end{aligned}
$$

so that

$$C_A = e_c C_A^{(c)} + e_{\bar b} C_A^{(\bar b)}.$$

For the tensor current,

$$
\begin{aligned}
C_B^{(c)} &= \frac{4i\,m_b m_c - 8i\,m_c^2}{m_b + m_c} - \frac{4i\,m_c^2 p_2}{(m_b + m_c)pq}, \\
C_B^{(\bar b)} &= -\frac{4i\,m_b m_c}{m_b + m_c} - \frac{4i\,m_b^2 p_2}{(m_b + m_c)pq},
\end{aligned}
$$

and

$$C_B = e_c C_B^{(c)} + e_{\bar b} C_B^{(\bar b)}.$$

The denominator-cancellation terms are stored in the corresponding output file. They should not be discarded automatically; they correspond to reduced two-denominator or contact contributions and must be treated in the dispersion/Borel analysis.

5 $B_{c1} \to B_c^* \gamma$: Levi-Civita Projection

For the vector final state a useful gauge-invariant structure is

$$V_{\mu\nu\rho} = p_\nu\, \epsilon_{\mu\rho p q} - (p \cdot q)\, \epsilon_{\mu\nu\rho p}.$$

It is transverse in the photon index for $q^2 = 0$. The FeynCalc checks are

$$V_{\mu\nu\rho} V^{\mu\nu\rho} = -4 p_2 (pq)^2, \qquad V_{\mu\nu\rho} \mathcal{P}^{\mu\nu\rho} = 1.$$

The projected J_A triangle cores are

$$
\begin{aligned}
C_{A,V}^{(c)} &= -\frac{4i\,m_b m_c}{p_2} + \frac{2i\,m_c^2}{p_2} + \frac{4i\,m_c^2}{pq}, \\
C_{A,V}^{(\bar b)} &= -\frac{2i\,m_b^2}{p_2} + \frac{4i\,m_b m_c}{p_2} + \frac{4i\,m_b^2}{pq}.
\end{aligned}
$$

The tensor-current vector cores are

$$
\begin{aligned}
C_{B,V}^{(c)} &= \frac{2i\,m_c}{m_b + m_c} + \frac{2i\,m_b^2 m_c - 4i\,m_b m_c^2 + 2i\,m_c^3}{(m_b + m_c)p_2} \\
&\quad - \frac{4i\,m_b m_c^2 - 4i\,m_c^3}{(m_b + m_c)pq} + \frac{2i\,m_c\,pq}{(m_b + m_c)p_2}, \\
C_{B,V}^{(\bar b)} &= \frac{2i\,m_b}{m_b + m_c} + \frac{-6i\,m_b^3 + 4i\,m_b^2 m_c + 2i\,m_b m_c^2}{(m_b + m_c)p_2} \\
&\quad - \frac{4i\,m_b^3 - 4i\,m_b^2 m_c}{(m_b + m_c)pq} + \frac{2i\,m_b\,pq}{(m_b + m_c)p_2}.
\end{aligned}
$$

Again,

$$C_{A,V} = e_c C_{A,V}^{(c)} + e_{\bar b} C_{A,V}^{(\bar b)}, \qquad C_{B,V} = e_c C_{B,V}^{(c)} + e_{\bar b} C_{B,V}^{(\bar b)}.$$

6 Controlled $B_c\gamma$ Pass

The script `Bc_gamma/scripts/bc_ps_complete_analysis.py` uses the compact $B_c\gamma$ triangle cores above and replaces the earlier common axial-vector normalization by physical two-point residues. We use the same central heavy-quark inputs and Borel window as the submitted B_c-mixing analysis,

$$m_b = 4.18 \text{ GeV}, \qquad m_c = 1.27 \text{ GeV}, \qquad \theta = 43.3^\circ,$$

more precisely $\theta_{\rm mean} = 43.295^\circ$ with $\sigma_\theta = 0.151^\circ$ and 16–84% interval $[43.147^\circ, 43.455^\circ]$ from the B_c-mixing Monte Carlo. The updated Monte Carlo samples this θ distribution.

$$M^2 = \{7, 8, 9\} \text{ GeV}^2, \qquad s_0 = \{53, 54, 55\} \text{ GeV}^2.$$

The pseudoscalar decay-constant benchmark is

$$f_{B_c} = 0.371 \pm 0.037 \text{ GeV}.$$

The physical axial-vector residues are extracted from the same two-point B_c-mixing basis. With $s_\theta = \sin\theta$ and $c_\theta = \cos\theta$,

$$\Pi_1 = s_\theta^2 \Pi_{AA} + 2s_\theta c_\theta \Pi_{AB} + c_\theta^2 \Pi_{BB}, \qquad \Pi_2 = c_\theta^2 \Pi_{AA} - 2s_\theta c_\theta \Pi_{AB} + s_\theta^2 \Pi_{BB},$$

and

$$f_i(M^2, s_0) = \left[\frac{e^{m_i^2/M^2} \Pi_i(M^2, s_0)}{m_i^2} \right]^{1/2}.$$

Using the perturbative heavy-heavy two-point spectral densities over the window above gives

$$f_1 = 0.264^{+0.005}_{-0.005} \text{ GeV}, \qquad f_2 = 0.722^{+0.016}_{-0.013} \text{ GeV}$$

from the 16–84% grid interval. This replaces the older common benchmark $f_{B_{c1}} = 0.373$ GeV.

Table 1: Updated $B_c\gamma$ results with physical two-point residues. The quoted intervals are the Monte Carlo 16–84% ranges over $m_b, m_c, M^2, s_0, \theta, f_{B_c}$, with f_1, f_2 recomputed from the two-point perturbative moments.

Channel	Status	Coupling/projection coefficient	Γ (keV)
$B_{c1}(6743) \to B_c\gamma$	controlled $B_c\gamma$	$+0.0315\,[0.0269, 0.0370]$ GeV^{-1}	$0.223\,[0.162, 0.307]$
$B_{c1}(6750) \to B_c\gamma$	controlled $B_c\gamma$	$-0.184\,[-0.205, -0.167]$ GeV^{-1}	$8.01\,[6.60, 9.90]$
$B_{c1}(6743) \to B_c^*\gamma$	diagonal vector pilot	$-0.0620\,[-0.0669, -0.0583]$	$46.3\,[41.0, 53.8]$
$B_{c1}(6750) \to B_c^*\gamma$	diagonal vector pilot	$+0.362\,[0.335, 0.401]$	$1.67\,[1.43, 2.04] \times 10^3$

The $B_c\gamma$ entries are the controlled hard-photon baseline from the present calculation. The shift relative to the first pilot is almost entirely normalization driven: f_1 is smaller than 0.373 GeV, while f_2 is larger. The $B_c^*\gamma$ entries should still be read as diagnostic numbers only. They are obtained from the gauge-invariant $V_{\mu\nu\rho}$ projection and a diagonal single-variable reduction. Before using them as final paper results one must match this coefficient to the physical $1^+ \to 1^-\gamma$ tensor basis and carry out the full double-Borel analysis, including denominator-cancellation or contact terms.

7 Comparison with Experiment and Other Theory

LHCb has observed a wide peaking structure in the $B_c^+\gamma$ mass spectrum with significance above seven standard deviations. A minimal two-peak description gives [1, 2]

$$M_1 = 6704.8 \pm 5.5 \pm 2.8 \pm 0.3 \text{ MeV}, \qquad M_2 = 6752.4 \pm 9.5 \pm 3.1 \pm 0.3 \text{ MeV},$$

where the errors are statistical, systematic, and due to the B_c^+ mass. No individual radiative partial width is measured. The natural widths are expected to be only a few hundred keV and are neglected in the experimental fit because they are small compared with the photon-energy resolution.

Table 2: Qualitative comparison with the current experimental information from LHCb [1, 2].

Quantity	Experiment	Present input/result
Observed $B_c(1P)^+$-like signal	yes, $> 7\sigma$ in $B_c^+\gamma$	channels studied explicitly
Lower visible peak	$6704.8 \pm 5.5 \pm 2.8 \pm 0.3$ MeV	input state 6743 MeV
Higher visible peak	$6752.4 \pm 9.5 \pm 3.1 \pm 0.3$ MeV	input state 6750 MeV
Partial widths	not measured	$B_c\gamma$: 0.223, 8.01 keV

The lower visible peak should not be identified one-to-one with the input 6743 MeV state without care. In the measured $B_c\gamma$ spectrum, decays through $B_c^*\gamma$, followed by an unreconstructed soft $B_c^* \to B_c\gamma$, can shift visible peaks by the unknown hyperfine splitting.

For comparison with other theory we should not compress different papers into a single range. Table 3 lists each quoted calculation separately. The entries are the values collected in Bondar–Milstein [4]; the original model papers are cited in the first column.

Table 3: Comparison with representative predictions for $B_{c1} \to B_c^{(*)}\gamma$. Widths are in keV.

Publication	$6743 \to B_c\gamma$	$6743 \to B_c^*\gamma$	$6750 \to B_c\gamma$	$6750 \to B_c^*\gamma$
Godfrey [5]	13	60	80	11
Ebert–Faustov–Galkin [6]	18.4	78.9	132	13.6
Fulcher [7]	32.4	75.6	127.8	26.2
Gershtein et al. [8]	11.6	77.8	131.1	8.1
T. Y. Li et al. [9]	16.6	49	66.6	10.5
Q. Li et al. [10]	35	70	74	40
X. J. Li et al. [11]	30.1	47.8	64	25.6
Eichten–Quigg [12]	9.9	62.5	92.3	7.5
Bondar–Milstein [4]	22	75	47	2
This work	$0.223[0.162, 0.307]$	$46.3[41.0, 53.8]$	$8.01[6.60, 9.90]$	$1.67[1.43, 2.04] \times 10^3$

Thus the $B_c\gamma$ baseline is already useful for comparison, while the $B_c^*\gamma$ channel must still be completed with the physical tensor-basis normalization and the full double-Borel/contact-term treatment.

8 Perturbative and Condensate Content

The $B_c\gamma$ numerical table above contains the leading hard-photon perturbative three-point baseline, normalized with perturbative two-point physical residues. In practical terms this means:

- the photon is emitted from the charm or anti-bottom heavy-quark line;
- both heavy propagators are free propagators in the radiative three-point OPE;
- the retained spectral density is the perturbative triangle hard contribution generated by the projected Dirac traces;
- no photon DA, $\chi\langle\bar{q}q\rangle$, or light-quark condensate contribution appears, because there is no light valence quark in B_c;
- gluon-condensate corrections from the background-gluon expansion of the heavy propagators are not yet included in the radiative three-point correlator;
- reduced two-denominator/contact terms from denominator cancellation still require a full double-Borel audit, especially in the vector $B_c^*\gamma$ channel.

Thus the present $B_c\gamma$ result is best described as a controlled leading perturbative hard-QCDSR baseline. A final publication-level B_c analysis should add, or explicitly bound, the heavy-heavy gluon-condensate corrections to the three-point function. The $B_c^*\gamma$ channel also requires the contact-term and tensor-normalization audit discussed above.

9 Dimension-4 Radiative Gluon-Condensate Workbench

The heavy-heavy B_c system has no photon DA or light-quark condensate contribution, but the background-gluon expansion of the heavy propagators generates possible dimension-4 gluon-condensate corrections to the radiative three-point function. To keep the calculation reproducible, the $\langle \alpha_s G^2/\pi \rangle$ part was split into small FeynCalc chunks instead of reducing one very large trace.

For the $B_c\gamma$ E1 structure the two hard-photon topologies are

$$S_c(k+q)\gamma_{\rm em}S_c(k)\gamma_5 S_b(k-p), \qquad S_c(k)\gamma_5 S_b(k-p)\gamma_{\rm em}S_b(k-p+q).$$

In each topology we considered

- one single-line insertion $S_Q^{(G^2)}$ on each heavy propagator segment;
- one open-gluon-pair insertion $S^{(G)}S^{(G)}$ for each pair of propagator segments.

Thus there are six single-line topologies and six open-pair topologies. Each topology is evaluated for the two basis currents J_A and J_B, giving 24 projected numerator rows in total.

The projection uses

$$S_{\mu\nu} = p_\nu q_\mu - (p\cdot q)g_{\mu\nu}, \qquad \mathcal{P}^{\mu\nu} = \frac{S^{\mu\nu}}{2(p\cdot q)^2},$$

so that the projected coefficient is the coefficient of the same E1 tensor used in the perturbative baseline. With the present conventions the audit is

class	current-split rows	projected terms
$S_Q^{(G^2)}$	12	221
$S^{(G)}S^{(G)}$	12	164
total	24	385

The scripts which generate these checks are

- `Bc_gamma/scripts/step3_ps_g2_topology_inventory.py`;
- `Bc_gamma/scripts/step3_ps_g2_single_line_cphoton.wl`;
- `Bc_gamma/scripts/step3_ps_g2_single_line_bbarphoton.wl`;
- `Bc_gamma/scripts/step3_ps_g2_open_pair_cphoton.wl`;
- `Bc_gamma/scripts/step3_ps_g2_open_pair_bbarphoton.wl`;
- `Bc_gamma/scripts/step3_ps_g2_projection_summary.py`;
- `Bc_gamma/scripts/step3_ps_g2_denominator_bookkeeping.py`.

The raw outputs are stored in `Bc_gamma/outputs/step3_ps_g2_*.txt` and the summary table is stored in `Bc_gamma/outputs/step3_ps_g2_projection_summary.csv`. The denominator manifest is stored in `Bc_gamma/outputs/step3_ps_g2_denominator_bookkeeping.csv`.

The denominator bookkeeping is most compact in Schwinger parameters. For the charm-line photon topology we use

$$a \leftrightarrow S_c(k+q), \qquad b \leftrightarrow S_c(k), \qquad r \leftrightarrow S_b(k-p), \qquad A = a+b+r.$$

After shifting the loop momentum the quadratic form is

$$\Phi_c = \frac{r(a+b)}{A}p^2 + \frac{2ar}{A}(p\cdot q) - (a+b)m_c^2 - rm_b^2 = \frac{rb}{A}p^2 + \frac{ar}{A}P^2 - (a+b)m_c^2 - rm_b^2.$$

For the anti-bottom-line photon topology we use

$$a \leftrightarrow S_c(k), \qquad b \leftrightarrow S_b(k-p), \qquad r \leftrightarrow S_b(k-p+q), \qquad A = a+b+r,$$

which gives

$$\Phi_{\bar b} = \frac{a(b+r)}{A}p^2 + \frac{2r(b+r)}{A}(p\cdot q) - am_c^2 - (b+r)m_b^2.$$

Equivalently, since $P^2 = p^2 + 2p\cdot q$,

$$\Phi_{\bar b} = \frac{(b+r)(a-r)}{A}p^2 + \frac{r(b+r)}{A}P^2 - am_c^2 - (b+r)m_b^2.$$

For $S_Q^{(G^2)}$, the selected propagator power is four and the other two propagator powers are one. For $S^{(G)}S^{(G)}$, the selected pair carries powers two and two, while the remaining propagator has power one.

This completes the numerator-projection stage of the radiative gluon-condensate workbench and attaches denominator powers to each projected row. It is not yet a numerical condensate correction to the decay width. The next step is to perform the double-dispersion/double-Borel reduction and integrate the result over the same Borel and continuum-threshold windows used for the perturbative baseline.

Conservative G^2 Size Screen

Before completing the full derivative-delta reduction, we performed a conservative size screen using the completed B_c-mixing OPE convergence table in the same numerical window,

$$M^2 = \{7,8,9\}\,\mathrm{GeV}^2, \qquad s_0 = \{53,54,55\}\,\mathrm{GeV}^2.$$

The script is `bc_ps_g2_screening_estimate.py`. The corresponding CSV and text outputs are stored in the `Bc_gamma/outputs` directory.

The screen defines

$$\Delta_{\mathrm{2pt}}^{G^2} = \max\left(\left|\frac{\Pi_{AA}^{G^2}}{\Pi_{AA}^{\mathrm{pert}}}\right|, \left|\frac{\Pi_{AB}^{G^2}}{\Pi_{AB}^{\mathrm{pert}}}\right|, \left|\frac{\Pi_{BB}^{G^2}}{\Pi_{BB}^{\mathrm{pert}}}\right| \right)$$

from the B_c-mixing calculation. Since $f \sim \sqrt{\Pi}$, the normalization shift is estimated as $\Delta_f \simeq \Delta_{\mathrm{2pt}}^{G^2}/2$. Taking the unknown radiative three-point G^2 shift to be of the same size gives the conservative envelope

$$\left|\frac{\delta g}{g}\right|_{\mathrm{env}} = \Delta_{\mathrm{2pt}}^{G^2} + \frac{1}{2}\Delta_{\mathrm{2pt}}^{G^2}, \qquad \left|\frac{\delta\Gamma}{\Gamma}\right|_{\mathrm{env}} \simeq 2\left|\frac{\delta g}{g}\right|_{\mathrm{env}}.$$

Numerically this gives a median conservative envelope $|\delta g/g|_{\mathrm{env}} = 0.058$ and $|\delta\Gamma/\Gamma|_{\mathrm{env}} = 0.115$, with maxima 0.128 and 0.257, respectively, over the grid.

Therefore the screen is not small enough to justify silently dropping G^2 at publication level. It does not prove that the radiative three-point G^2 correction is as large as the envelope; it shows that a few-percent amplitude effect is plausible in the same heavy-heavy OPE window. The safe statement is that the present widths are leading perturbative hard-QCDSR baselines until the explicit radiative G^2 reduction is completed or bounded more sharply.

10 Reproduction Checklist for the Present Pilot

The current numbers can be reproduced in the following order.

1. Run `Bc_gamma/scripts/step1_hard_photon_numerators.wl`. This computes the raw FeynCalc traces for photon emission from the charm line and from the anti-bottom line.

2. Run `Bc_gamma/scripts/step2_ps_e1_triangle_reduction.wl`. This projects the $B_c\gamma$ traces onto $S_{\mu\nu} = p_\nu q_\mu - (p \cdot q)g_{\mu\nu}$ and checks $S_{\mu\nu}\mathcal{P}^{\mu\nu} = 1$.

3. Run `Bc_gamma/scripts/step2_vec_epsilon_A_reduction.wl` and `Bc_gamma/scripts/step2_vec_epsilon_B_reduction.wl`. These produce the diagnostic $B_c^*\gamma$ cores and check the normalization of $V_{\mu\nu\rho}$.

4. Insert the compact projected cores into `Bc_gamma/scripts/bc_ps_complete_analysis.py`. The script uses the diagonal calibration $M_1^2 = M_2^2 = 2M^2$, with $M^2 = \{7, 8, 9\}\,\mathrm{GeV}^2$ and $s_0 = \{53, 54, 55\}\,\mathrm{GeV}^2$.

5. Extract f_1 and f_2 from the diagonalized two-point perturbative moments and then sample $m_b, m_c, M^2, s_0, \theta$, and f_{B_c} in the Monte Carlo.

6. Rotate the basis amplitudes with $\widehat{\Pi}_{6743} = \sin\theta\,\widehat{\Pi}_A + \cos\theta\,\widehat{\Pi}_B$ and $\widehat{\Pi}_{6750} = \cos\theta\,\widehat{\Pi}_A - \sin\theta\,\widehat{\Pi}_B$.

7. Convert the pseudoscalar final-state coupling to the width using

$$\Gamma(1^+ \to 0^-\gamma) = \frac{\alpha_{\rm em}}{3} g^2 \left(\frac{m_i^2 - m_{B_c}^2}{2m_i}\right)^3.$$

At this stage the $B_c\gamma$ entries are the controlled hard-photon baseline. The $B_c^*\gamma$ entries are useful diagnostics, but they should not be treated as final publication numbers until the physical $1^+ \to 1^-\gamma$ tensor basis, the full double-Borel treatment, and the reduced-denominator/contact terms have been completed.

11 Next Step

The double-dispersion variables satisfy

$$P^2 = p^2 + 2p \cdot q, \qquad pq = \frac{P^2 - p^2}{2}.$$

For $B_c\gamma$, the structure is close to the strange-sector E1 case, but with two heavy masses. For $B_c^*\gamma$, the additional $1/p_2$ and pq/p_2 structures mean that the double dispersion should be written explicitly before numerical Borel analysis. A diagonal single-variable replacement can be useful as a calibration check, but it should not be the final derivation for the vector final state.

Double-dispersion denominator

For the photon emitted from the charm line, the three denominators are

$$D_1 = (k+q)^2 - m_c^2, \qquad D_2 = k^2 - m_c^2, \qquad D_3 = (k-p)^2 - m_b^2.$$

With Feynman parameters x, y, z, $x+y+z=1$, the combined denominator is

$$xD_1 + yD_2 + zD_3 = \ell^2 - \Delta_c,$$

where

$$\Delta_c = (1-z)m_c^2 + zm_b^2 - zy\,p^2 - xz\,P^2.$$

For photon emission from the $\bar{b}$ line, using the momentum routing in the FeynCalc scripts,

$$D_1 = k^2 - m_c^2, \qquad D_2 = (k-p)^2 - m_b^2, \qquad D_3 = (k-p+q)^2 - m_b^2,$$

and the analogous reduction gives, with $x+y+z=1$,

$$\Delta_{\bar{b}} = xm_c^2 + (1-x)m_b^2 - x(1-x)p^2 + xz(P^2 - p^2).$$

Equivalently one may relabel the Feynman parameters after choosing a different loop momentum. These forms make clear why the true radiative problem is a double-dispersion problem in p^2 and P^2, not simply the two-point mixing correlator with an extra charge factor.

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